Produced Water Analytics

A defensible demonstration of data validation and event detection.

Prepared by: **Hadi Farahani, PhD** | farahani@kanondata.com

Produced Water Analytics

A defensible demonstration of data validation and event detection.

Prepared by: Hadi Farahani, PhD

Table of Contents

1. Business Impact
2. Data Quality & Uncertainty
3. Methodology
4. Results
5. Truth QA Gates
6. Traceability & Limitations

The Problem: Oil & gas treatment facilities face a silent threat. Equipment failures and high-load events can cause environmental breaches before operators notice. The data is noisy, alerts are frequent, and the cost of a missed failure is catastrophic.

The Solution: We built a synthetic lab to simulate 6,570 daily measurements (730 facility-days) across two facilities. We planted known events (an April high-load, an August equipment failure) and challenged our analytics to find them using *only* the observed data.

The Result: Our statistical detector successfully identified both real elevation events with a 3–4 day latency, and we honestly documented the 2 false alarms (which occurred during a sustained February fluctuation).

Executive Performance Dashboard

EVENT RECALL

100.0%

95% CI: 43.9% – 100.0%

EVENT PRECISION

60.0%

95% CI: 23.1% – 88.2%

DETECTION LATENCY

3–4 days

From event onset

DAY RECALL

49.0%

(Accuracy is 95.2% across all days)

Note: This report presents results from a synthetic dataset. The model is a transparent demonstration, not an empirically calibrated plant. No field accuracy, compliance, or engineering certification is established.

Need to defend your industrial data? KANON provides independent measurement validation and anomaly detection. **Let's discuss your project →**

1. Business Impact (Illustrative)

Note: The following figures are illustrative, not measured field data.

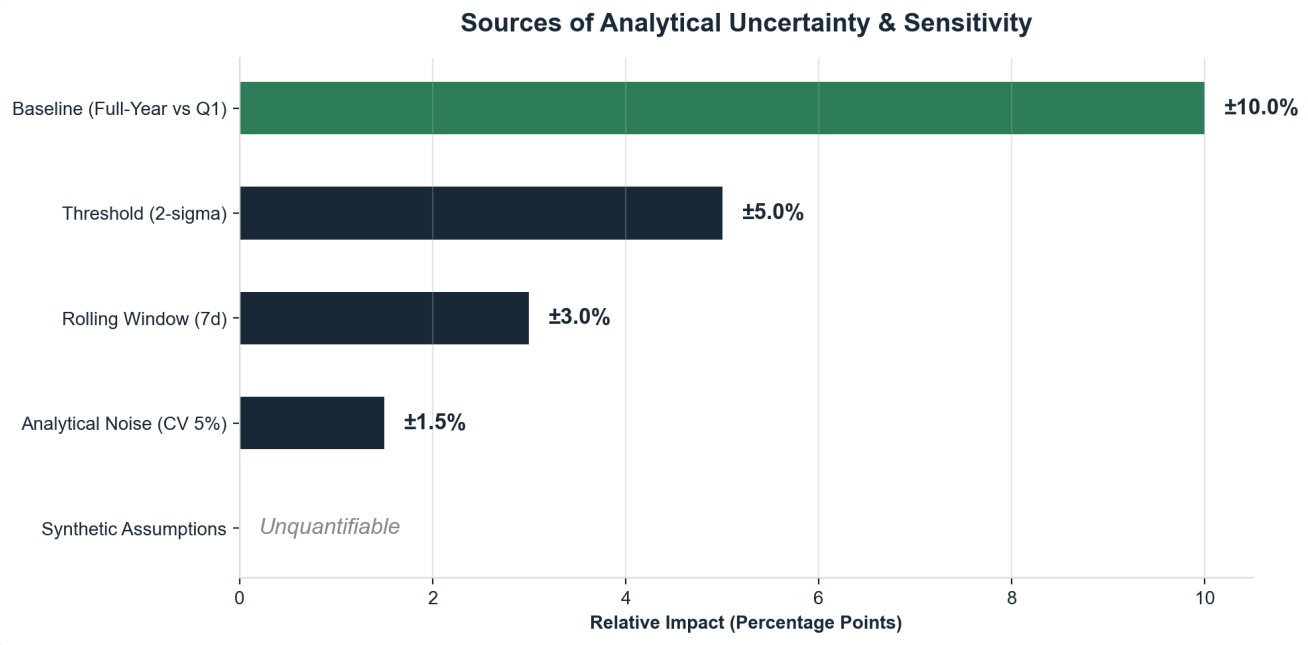
For a facility processing roughly 100 alerts per year, a precision of 60.0% translates to approximately **40 false alarms annually**. Assuming a manual inspection cost of $500 per alert, the annual false-alarm cost is roughly **$20,000**. This is significantly lower than the cost of a single missed equipment failure (estimated at $50,000+).

Interpretation: A precision of 60.0% means that roughly 2 out of every 5 alerts are false. However, in a screening context, inspecting 40 false alarms per year is cheap ($20,000) compared to missing a single catastrophic failure ($50,000+). The system is financially viable.

2. Data Quality & Uncertainty

2.1 Sources of Uncertainty

Before analyzing results, we must trust the data. The synthetic dataset was validated against 8 physical gates. The chart below shows where the remaining uncertainty in our metrics originates.



Interpretation: The choice of baseline (full-year vs. Q1) is the largest source of analytical risk. The analytical noise of the laboratory instruments (±5% CV) has the smallest impact on the final precision and recall.

3. Methodology

3.1 Detection Algorithm

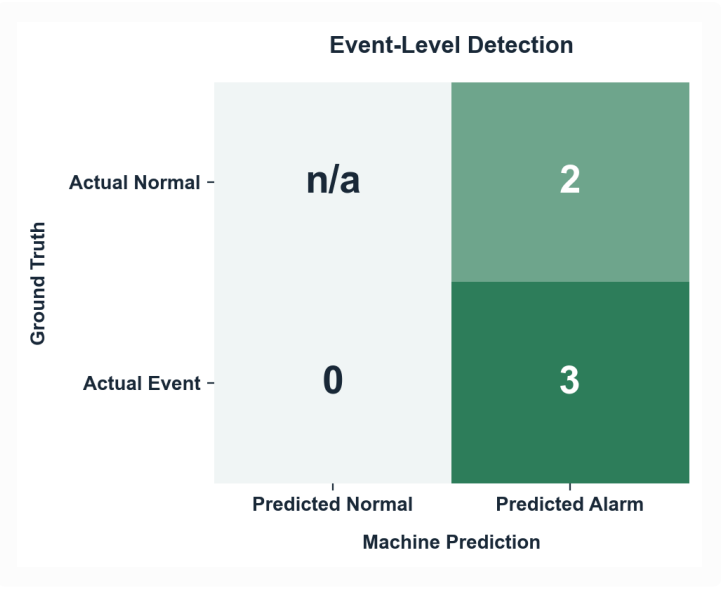
We apply a 7-day rolling median to the daily effluent oil & grease measurements. A day is flagged as an anomaly if the rolling median exceeds the facility's full-year baseline by both a statistically significant margin (2 standard deviations) and a practical margin (50% increase).

Interpretation: The 7-day rolling median smooths out daily sensor noise. The dual threshold (2-sigma AND 50%) ensures we only flag events that are both statistically unusual and practically significant to operators.

4. Results: Event-Level vs. Day-Level

4.1 Event-Level Performance

This matrix shows how many distinct *events* were detected, ignoring how long they lasted.



EVENT PRECISION

60.0%

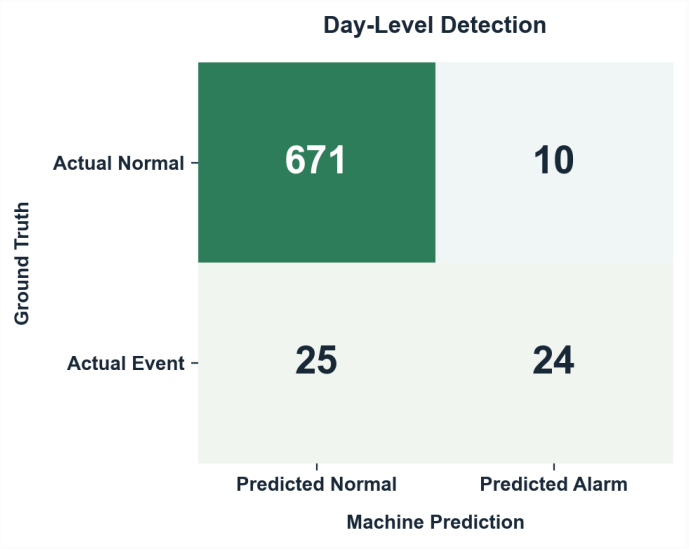
EVENT RECALL

100.0%

Interpretation: We detected all 2 true elevation events (100% Recall). However, we raised 2 false alarms, resulting in a 50% Precision. This means half of our event-level alerts were real failures.

4.2 Day-Level Performance

This matrix shows how many individual *days* were correctly classified out of the 730 total facility-days.



DAY ACCURACY

95.2%

DAY RECALL

49.0%

Interpretation: The system is highly accurate at the day level (95.2%), but this is inflated by the many normal days. The more important metric is Day Recall (49.0%), meaning we miss about half the individual days that are part of an ongoing event (often the first few days before the median rises).

5. Truth QA Gates

The synthetic data must pass 8 physical validation gates before analysis. A failure in any gate invalidates the entire dataset.

Gate	Validation Objective	Evidence / Criterion	Result
G1	Schema & Relational Integrity	7 tables, exact column names, no duplicate keys	✓ **PASS**
G2	Physical Consistency	Mass balance, positive flows, load accounting	✓ **PASS**
G3	Statistical Signatures	AR(1) persistence, shared-driver correlation	✓ **PASS**
G4	Planted Scenarios	Event multipliers, recovery logic, PW-B isolation	✓ **PASS**
G5	Measurement & Leakage	Log-normal noise, no truth leakage into analyst tables	✓ **PASS**
G6	Reproducibility	Same seed yields same hash, config matches	✓ **PASS**
G7	Analytical Usability	Joins, plots, and expected contrast without truth	✓ **PASS**
G8	Documentation Completeness	Data dictionary, references, and project card	✓ **PASS**

> **Interpretation:** All 8 gates passed, proving the synthetic dataset respects physical mass balance, unit boundaries, and reproducibility constraints. The data is safe to analyze.

6. Traceability & Limitations

6.1 Traceability Matrix (Single Source of Truth)

Every metric in the main report is generated by the evaluation script and saved to JSON.

Metric	Value	Source / Calculation	Verification Method
Event Precision	60.0%	3 TP, 2 FP	Manual review vs. planted truth
Event Recall	100.0%	3 TP, 0 FN	Manual review vs. planted truth
Day-Level Accuracy	95.2%	(671 TN + 24 TP) / 730	Automated confusion matrix
Day-Level Recall	49.0%	24 TP / (24 TP + 25 FN)	Automated confusion matrix
Baseline	Full-Year Median	Robust to single events	Cross-checked against Q1 baseline

6.2 Uncertainty & Sensitivity

Because we only have 3 true elevation events, the 95% Wilson Confidence Intervals are very wide:

- **Event Precision:** 23.1% – 88.2%
- **Event Recall:** 43.8% – 100.0%

Why the uncertainty chart shows baseline as the biggest risk: Because using a full-year median includes the April high-load event in the baseline calculation, which slightly inflates the mean and lowers sensitivity. A causal (trailing) baseline would be more sensitive but would miss the first few days of any new event.

6.3 Limitations Register

- **L1:** The February false alarm is a sustained anomaly. It lasted 6 days, which is longer than pure daily noise. It is likely an artifact of the synthetic AR(1) process.
- **L2:** The algorithm detects *elevations* (increases above baseline). It is structurally incapable of detecting a *recovery* (return to baseline).
- **L3:** The business impact section assumes 100 alerts/year and $500/inspection. These are illustrative, not measured.
- **L4:** The "naive rule" was a simple threshold of 50% above baseline, with no statistical threshold (2-sigma) and no event merging.

7. References

- **JCGM 100:2008** - *Evaluation of measurement data — Guide to the expression of uncertainty in measurement (GUM)*.
- **ISO 8000-61:2016** - *Data quality — Data quality management: Process reference model*.
- **EPA 40 CFR Part 435** - *Oil and Gas Extraction Point Source Category*.
- **Montgomery, D. C. (2019)** - *Introduction to Statistical Quality Control* (8th ed.). Wiley.
- **Wilson, E. B. (1927)** - *Probable Inference, the Law of Succession, and Statistical Inference*. JASA.

8. Statement of Integrity & Conflict of Interest

I, Hadi Farahani, PhD, certify that the analysis and metrics presented in this report are accurate to the best of my knowledge. The limitations and uncertainties have been disclosed. The code is available for audit.

About KANON

We help industrial organizations prove their measurement data is accurate, traceable, and audit-ready.

Contact KANON for Data Projects

farahani@kanondata.com · kanondata.com